

Halogen compounds

A Level Chemistry

Halogenoalkanes

A **halogenoalkane** 卤代烷 is an alkane with one or more halogen atoms in place of hydrogen (a C-X bond, where X is a halogen).

Making halogenoalkanes

- **free-radical substitution** 自由基取代 of an alkane with Cl₂ or Br₂ in ultraviolet light.
- **electrophilic addition** 亲电加成 of an alkene with a halogen X₂ or a hydrogen halide HX.
- substitution of an **alcohol** 醇, for example by HX, by PCl₅, by PCl₃ with heat, or by SOCl₂.

Three classes

A halogenoalkane is **primary** 伯, **secondary** 仲 or **tertiary** 叔, depending on how many carbon atoms are joined to the carbon that holds the halogen (one, two or three).

Nucleophilic substitution

The C-X bond is polar, so the carbon is slightly positive. A **nucleophilic substitution** 亲核取代 happens when a **nucleophile** 亲核试剂 (a lone-pair species) attacks that carbon and replaces the halogen.

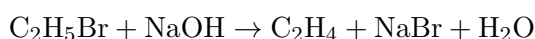
Reagent and conditions	Product
NaOH(aq), heat	an alcohol
KCN in ethanol, heat	a nitrile 腈 (adds one carbon to the chain)
NH ₃ in ethanol, heated under pressure	an amine 胺

To **identify the halogen**, warm the halogenoalkane with **silver nitrate** 硝酸银 in ethanol. A silver halide **precipitate** 沉淀 forms, and its colour shows which halogen is present (white AgCl, cream AgBr, yellow AgI).

Elimination

The same halogenoalkane can instead undergo **elimination** 消去 to form an **alkene** 烯烃. The conditions decide which reaction wins:

- NaOH in **water** → nucleophilic substitution → an alcohol.
- NaOH in **ethanol**, heated → elimination → an alkene.



The S_N1 and S_N2 mechanisms

Nucleophilic substitution can follow two routes:

- **S_N2**: one step. The nucleophile attacks at the same time as the halogen leaves, passing through a crowded **transition state** 过渡态 where both are half-bonded. The rate depends on both the halogenoalkane and the nucleophile.
- **S_N1**: two steps. First the C–X bond breaks to give a **carbocation** 碳正离子; then the nucleophile attacks it. The rate depends only on the halogenoalkane.

Alkyl groups push electrons towards the positive carbon (the **inductive effect** 诱导效应), so they stabilise the carbocation. This is why:

- **primary** halogenoalkanes mostly react by S_N2.
- **tertiary** halogenoalkanes mostly react by S_N1 (their carbocation is well stabilised).
- **secondary** halogenoalkanes use a mixture of the two.

Different reactivities

How fast a halogenoalkane reacts depends on the strength of the C–X bond, measured by its **bond energy** 键能. The C–I bond is the weakest, so iodoalkanes react fastest; the C–Cl bond is the strongest of the three, so chloroalkanes react slowest. So when tested with silver nitrate, an iodoalkane gives its precipitate first —its higher **reactivity** 反应活性 comes from the weaker C–X bond.